

2

Improper Integrals

1. Find each of the following limits.

$$(a) \lim_{t \rightarrow 1^-} \sqrt{t} = \sqrt{1} = 1$$

$$(b) \lim_{t \rightarrow 1^-} \sin^{-1} t = \sin^{-1} 1 = \frac{\pi}{2}$$

$$(c) \lim_{t \rightarrow 1^+} \ln|1 - t| = -\infty$$

$$(d) \lim_{x \rightarrow \infty} (\tan^{-1} x)^2 = \left(\lim_{x \rightarrow \infty} \tan^{-1} x \right)^2 = \left(\frac{\pi}{2} \right)^2 = \frac{\pi^2}{4}$$

$$(e) \lim_{x \rightarrow \infty} \tan^{-1}(\sqrt{x^2 - 1}) = \lim_{y \rightarrow \infty} \tan^{-1} y = \frac{\pi}{2}$$

$$(f) \lim_{x \rightarrow \infty} \frac{x+2}{x+3} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(x+2)}{\frac{d}{dx}(x+3)} = \lim_{x \rightarrow \infty} \frac{1}{1} = 1$$

$$(g) \lim_{x \rightarrow \infty} \ln \left| \frac{x+2}{x+3} \right| = \lim_{y \rightarrow 1} \ln|y| = \ln 1 = 0$$

$$(h) \lim_{x \rightarrow \infty} e^{-x^2} = \lim_{x \rightarrow \infty} \frac{1}{e^{x^2}} = 0 \text{ c.f. Limit Rule \#3}$$

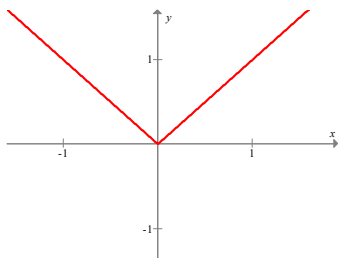
$$(i) \lim_{x \rightarrow -\infty} x e^x = \lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} = \lim_{y \rightarrow \infty} \frac{-y}{e^y} = \lim_{y \rightarrow \infty} \frac{\frac{d}{dy}(-y)}{\frac{d}{dy}(e^y)} = \lim_{y \rightarrow \infty} \frac{-1}{e^y} = 0$$

$$(j) \lim_{x \rightarrow \infty} e^{-x} \sin x = \lim_{x \rightarrow \infty} \frac{\sin x}{e^x} = 0 \text{ since } \frac{-1}{e^x} \leq \frac{\sin x}{e^x} \leq \frac{1}{e^x} \text{ and } \lim_{x \rightarrow \infty} \frac{-1}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

2. A function f is said to be **continuous** at $x = x_0$ if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. Sketch the graph of each of the following functions, and determine whether it is continuous at the specified point by calculating its left and right limits at that point.



(a) $f(x) = |x|$ at $x = 0$

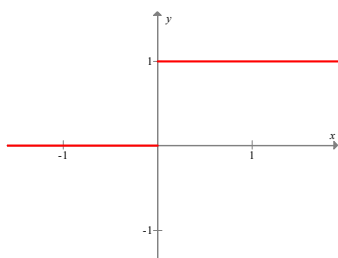


$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

$$\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$$

Since $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$,
 $f(x)$ is continuous at $x = 0$

(b) $f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$ at $x = 0$

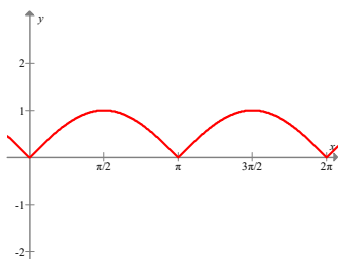


$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 0 = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 1 = 1$$

Since $\lim_{x \rightarrow 0} f(x)$ does not exist,
 $f(x)$ is not continuous at $x = 0$

(c) $f(x) = |\sin x|$ at $x = \pi$



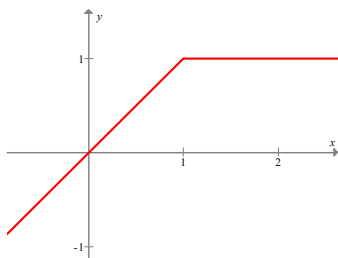
$$\lim_{x \rightarrow \pi^-} |\sin x| = \lim_{x \rightarrow \pi^-} \sin x = \sin \pi = 0$$

$$\lim_{x \rightarrow \pi^+} |\sin x| = \lim_{x \rightarrow \pi^+} (-\sin x) = -\sin \pi = 0$$

Since $\lim_{x \rightarrow \pi} f(x) = 0 = f(\pi)$,
 $f(x)$ is continuous at $x = \pi$



(d) $f(x) = \begin{cases} x & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$ at $x = 1$



$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 1 = 1$$

Since $\lim_{x \rightarrow 1} f(x) = 1 = f(1)$,
 $f(x)$ is continuous at $x = 1$

3. The **derivative** $f'(x)$ of a function f is defined as $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$.

Without applying l'Hôpital rule, use the definition above to find the derivative of each of the following functions.

(a) $f(x) = \sin x$

$$\begin{aligned} f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{\sin(x + \Delta x) - \sin x}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\sin x \cos \Delta x + \cos x \sin \Delta x - \sin x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \left(\sin x \frac{\cos \Delta x - 1}{\Delta x} + \cos x \frac{\sin \Delta x}{\Delta x} \right) \\ &= \sin x \lim_{\Delta x \rightarrow 0} \left(\frac{\cos \Delta x - 1}{\Delta x} \right) + \cos x \lim_{\Delta x \rightarrow 0} \left(\frac{\sin \Delta x}{\Delta x} \right) = \cos x \end{aligned}$$

(b) $f(x) = \cos x$

$$\begin{aligned} f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{\cos(x + \Delta x) - \cos x}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\cos x \cos \Delta x - \sin x \sin \Delta x - \cos x}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \left(\cos x \frac{\cos \Delta x - 1}{\Delta x} - \sin x \frac{\sin \Delta x}{\Delta x} \right) = -\sin x \end{aligned}$$



(c) $f(x) = |x|$ at $x = 0$

$$\begin{aligned} f'(0) &= \lim_{\Delta x \rightarrow 0} \frac{|0 + \Delta x| - |0|}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{|\Delta x|}{\Delta x} \\ \lim_{\Delta x \rightarrow 0^+} \frac{|\Delta x|}{\Delta x} &= \lim_{\Delta x \rightarrow 0^+} \frac{\Delta x}{\Delta x} = 1 \text{ but } \lim_{\Delta x \rightarrow 0^-} \frac{|\Delta x|}{\Delta x} = \lim_{\Delta x \rightarrow 0^-} \frac{-\Delta x}{\Delta x} = -1 \\ \text{So, } f'(0) &= \lim_{\Delta x \rightarrow 0} \frac{|\Delta x|}{\Delta x} \text{ does not exist} \end{aligned}$$



(d) $f(x) = \begin{cases} x & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$ at $x = 1$

$$\begin{aligned} f'(1) &= \lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - 1}{\Delta x} \\ \lim_{\Delta x \rightarrow 0^+} \frac{f(1 + \Delta x) - 1}{\Delta x} &= \lim_{\Delta x \rightarrow 0^+} \frac{1 - 1}{\Delta x} = 0 \\ \lim_{\Delta x \rightarrow 0^-} \frac{f(1 + \Delta x) - 1}{\Delta x} &= \lim_{\Delta x \rightarrow 0^-} \frac{1 + \Delta x - 1}{\Delta x} = \lim_{\Delta x \rightarrow 0^-} \frac{\Delta x}{\Delta x} = 1 \\ \text{So, } f'(1) &= \lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - 1}{\Delta x} \text{ does not exist} \end{aligned}$$

4. Calculate each of the following improper integrals of the 1st kind.

$$(a) \int_0^{\infty} \frac{1}{x^2 + 1} dx = \lim_{N \rightarrow \infty} \int_0^N \frac{1}{x^2 + 1} dx = \lim_{N \rightarrow \infty} \left[\tan^{-1} x \right]_0^N = \lim_{N \rightarrow \infty} \tan^{-1} N - \tan^{-1} 0 = \frac{\pi}{2}$$

$$(b) \int_1^{\infty} \frac{1}{x^{1.001}} dx = \lim_{N \rightarrow \infty} \int_1^N \frac{1}{x^{1.001}} dx = \lim_{N \rightarrow \infty} \left[\frac{x^{-0.001}}{-0.001} \right]_1^N \\ = -1000 \left(\lim_{N \rightarrow \infty} N^{-0.001} - 1^{-0.001} \right) = 1000$$

$$(c) \int_2^{\infty} \frac{1}{(x-1)^2} dx = \lim_{N \rightarrow \infty} \int_2^N \frac{1}{(x-1)^2} dx = \lim_{N \rightarrow \infty} \left[\frac{(x-1)^{-1}}{-1} \right]_2^N \\ = - \left(\lim_{N \rightarrow \infty} (N-1)^{-1} - (2-1)^{-1} \right) = 1$$

$$(d) \int_0^{\infty} \frac{16 \tan^{-1} x}{x^2 + 1} dx = 16 \lim_{N \rightarrow \infty} \int_0^N \tan^{-1} x \left(\frac{1}{x^2 + 1} dx \right) = 16 \lim_{N \rightarrow \infty} \left[\frac{(\tan^{-1} x)^2}{2} \right]_0^N \\ = 8 \left(\lim_{N \rightarrow \infty} (\tan^{-1} N)^2 - (\tan^{-1} 0)^2 \right) = 8 \left(\frac{\pi^2}{4} \right) = 2\pi^2$$

$$(e) \int_2^{\infty} \frac{2}{t^2 - 1} dt = 2 \lim_{N \rightarrow \infty} \int_2^N \frac{1}{t^2 - 1} dt = 2 \lim_{N \rightarrow \infty} \left[-\frac{1}{2} \ln \left| \frac{1+t}{1-t} \right| \right]_2^N \\ = - \left(\lim_{N \rightarrow \infty} \ln \left| \frac{1+N}{1-N} \right| - \ln \left| \frac{1+2}{1-2} \right| \right) = - \left(\lim_{y \rightarrow -1} \ln |y| - \ln |-3| \right) = \ln 3$$

$$(f) \int_{-1}^{\infty} \frac{1}{x^2 + 5x + 6} dx = \lim_{N \rightarrow \infty} \int_{-1}^N \frac{1}{(x+2.5)^2 - 0.25} dx \\ = \lim_{N \rightarrow \infty} \left[-\frac{1}{1} \ln \left| \frac{0.5 + (x+2.5)}{0.5 - (x+2.5)} \right| \right]_{-1}^N \\ = - \left(\lim_{N \rightarrow \infty} \ln \left| \frac{3+N}{-2-N} \right| - \ln \left| \frac{2}{-1} \right| \right) = - \left(\lim_{y \rightarrow -1} \ln |y| - \ln 2 \right) = \ln 2$$

$$(g) \int_{-\infty}^0 x e^x dx = \lim_{N \rightarrow -\infty} \int_N^0 x e^x dx = \lim_{N \rightarrow -\infty} \left[x e^x - e^x \right]_N^0 \\ = \lim_{N \rightarrow -\infty} \left((0 - e^0) - (N e^N - e^N) \right) = -1 - \lim_{N \rightarrow -\infty} (-N e^{-N} - e^{-N}) \\ = -1 + \lim_{N \rightarrow \infty} \left(\frac{N+1}{e^N} \right) = -1 + \lim_{N \rightarrow \infty} \left(\frac{1}{e^N} \right) = -1$$

$$\begin{aligned}
 \text{(h)} \quad \int_0^{\infty} 2e^{-x} \sin x \, dx &= 2 \lim_{N \rightarrow \infty} \int_0^N e^{-x} \sin x \, dx = 2 \lim_{N \rightarrow \infty} \left[-\frac{1}{2} e^{-x} (\sin x + \cos x) \right]_0^N \\
 &= - \lim_{N \rightarrow \infty} (e^{-N} (\sin N + \cos N) - e^0 (\sin 0 + \cos 0)) \\
 &= - \lim_{N \rightarrow \infty} \left(\frac{\sin N}{e^N} \right) - \lim_{N \rightarrow \infty} \left(\frac{\cos N}{e^N} \right) + 1 = 1 \text{ by Squeeze Theorem}
 \end{aligned}$$

$$\begin{aligned}
 \text{(i)} \quad \int_{-\infty}^{\infty} 2xe^{-x^2} \, dx &= \lim_{M \rightarrow -\infty} \int_M^0 2xe^{-x^2} \, dx + \lim_{N \rightarrow \infty} \int_0^N 2xe^{-x^2} \, dx \\
 &= \lim_{M \rightarrow -\infty} \left[-e^{-x^2} \right]_M^0 + \lim_{N \rightarrow \infty} \left[-e^{-x^2} \right]_0^N \\
 &= \lim_{M \rightarrow -\infty} (-e^0 + e^{-M^2}) + \lim_{N \rightarrow \infty} (-e^{-N^2} + e^0) = -1 + 0 + 0 + 1 = 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(j)} \quad \int_{-\infty}^{\infty} \frac{x}{(x^2 + 4)^{3/2}} \, dx &= \frac{1}{2} \left(\lim_{M \rightarrow -\infty} \int_M^0 \frac{2x}{(x^2 + 4)^{3/2}} \, dx + \lim_{N \rightarrow \infty} \int_0^N \frac{2x}{(x^2 + 4)^{3/2}} \, dx \right) \\
 &= \frac{1}{2} \left(\lim_{M \rightarrow -\infty} \left[\frac{(x^2 + 4)^{-1/2}}{-\frac{1}{2}} \right]_M^0 + \lim_{N \rightarrow \infty} \left[\frac{(x^2 + 4)^{-1/2}}{-\frac{1}{2}} \right]_0^N \right) \\
 &= - \left(\lim_{M \rightarrow -\infty} \left(\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{M^2 + 4}} \right) \right) + \lim_{N \rightarrow \infty} \left(\frac{1}{\sqrt{N^2 + 4}} - \frac{1}{\sqrt{4}} \right) \\
 &= - \left(\frac{1}{2} - 0 + 0 - \frac{1}{2} \right) = 0
 \end{aligned}$$

5. Calculate each of the following improper integrals of the 2nd kind.

$$(a) \int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} dx = \lim_{t \rightarrow 0^+} \left[\frac{x^{1/2}}{1/2} \right]_t^1 = 2 \left(\sqrt{1} - \lim_{t \rightarrow 0^+} \sqrt{t} \right) = 2$$

$$(b) \int_0^1 \frac{1}{\sqrt{1-x^2}} dx = \lim_{t \rightarrow 1^-} \int_0^t \frac{1}{\sqrt{1-x^2}} dx = \lim_{t \rightarrow 1^-} \left[\sin^{-1} x \right]_0^t = \lim_{t \rightarrow 1^-} \sin^{-1} t - \sin^{-1} 0 = \frac{\pi}{2}$$

$$(c) \int_0^1 \frac{x+1}{\sqrt{x^2+2x}} dx = \frac{1}{2} \lim_{t \rightarrow 0^+} \int_t^1 \frac{2x+2}{\sqrt{x^2+2x}} dx = \frac{1}{2} \lim_{t \rightarrow 0^+} \left[\frac{(x^2+2x)^{1/2}}{1/2} \right]_t^1 \\ = \lim_{t \rightarrow 0^+} \left(\sqrt{1+2} - \sqrt{t^2+2t} \right) = \sqrt{3}$$

$$(d) \int_0^1 x \ln x dx = \lim_{t \rightarrow 0^+} \int_t^1 x \ln x dx = \lim_{t \rightarrow 0^+} \left[\frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 \right]_t^1 \\ = \lim_{t \rightarrow 0^+} \left(\left(\frac{1}{2} \ln 1 - \frac{1}{4} \right) - \left(\frac{1}{2} t^2 \ln t - \frac{1}{4} t^2 \right) \right) = -\frac{1}{4} - \frac{1}{2} \lim_{t \rightarrow 0^+} t^2 \ln t \\ = -\frac{1}{4} - \frac{1}{2} \lim_{t \rightarrow 0^+} \frac{\ln t}{t^{-2}} = -\frac{1}{4} - \frac{1}{2} \lim_{t \rightarrow 0^+} \frac{t^{-1}}{-2t^{-3}} = -\frac{1}{4} + \frac{1}{4} \lim_{t \rightarrow 0^+} t^2 = -\frac{1}{4}$$

$$(e) \int_0^1 \frac{4x}{\sqrt{1-x^4}} dx = 2 \lim_{t \rightarrow 1^-} \int_0^t \frac{2x}{\sqrt{1-(x^2)^2}} dx = 2 \lim_{t \rightarrow 1^-} \left[\sin^{-1}(x^2) \right]_0^t \\ = 2 \left(\lim_{t \rightarrow 1^-} \sin^{-1} t^2 - \sin^{-1} 0 \right) = \pi$$

$$(f) \int_0^{\pi/2} (\sec x - \tan x) dx = \lim_{t \rightarrow \pi/2^-} \int_0^t (\sec x - \tan x) dx \\ = \lim_{t \rightarrow \pi/2^-} \left[\ln|\sec x + \tan x| + \ln|\cos x| \right]_0^t \\ = \lim_{t \rightarrow \pi/2^-} \left[\ln|1 + \sin x| \right]_0^t \\ = \lim_{t \rightarrow \pi/2^-} \ln|1 + \sin t| - \ln|1 + \sin 0| \\ = \ln \left| 1 + \sin \frac{\pi}{2} \right| - \ln 1 = \ln 2$$

$$\begin{aligned}
 \text{(g)} \quad \int_{-1}^1 \frac{1}{x^{2/3}} dx &= \lim_{s \rightarrow 0^-} \int_{-1}^s x^{-2/3} dx + \lim_{t \rightarrow 0^+} \int_t^1 x^{-2/3} dx \\
 &= \lim_{s \rightarrow 0^-} \left[\frac{x^{1/3}}{1/3} \right]_{-1}^s + \lim_{t \rightarrow 0^+} \left[\frac{x^{1/3}}{1/3} \right]_t^1 \\
 &= 3 \left(\lim_{s \rightarrow 0^-} s^{1/3} - (-1)^{1/3} + 1^{1/3} - \lim_{t \rightarrow 0^+} t^{1/3} \right) = 3(0 + 1 + 1 - 0) = 6
 \end{aligned}$$

$$\begin{aligned}
 \text{(h)} \quad \int_1^2 \frac{1}{1-x} dx &= \lim_{t \rightarrow 1^+} \int_t^2 \frac{1}{1-x} dx \\
 &= \lim_{t \rightarrow 1^+} \left[-\ln|1-x| \right]_t^2 \\
 &= -\ln|-1| + \underbrace{\lim_{t \rightarrow 1^+} \ln|1-t|}_{\text{divergent}}
 \end{aligned}$$

$$\therefore \int_1^2 \frac{1}{1-x} dx \text{ is divergent}$$

$$\begin{aligned}
 \text{(i)} \quad \int_0^2 \frac{1}{\sqrt{|x-1|}} dx &= \lim_{s \rightarrow 1^-} \int_0^s \frac{1}{\sqrt{1-x}} dx + \lim_{t \rightarrow 1^+} \int_t^2 \frac{1}{\sqrt{x-1}} dx \\
 &= \lim_{s \rightarrow 1^-} \left[-\frac{\sqrt{1-x}}{1/2} \right]_0^s + \lim_{t \rightarrow 1^+} \left[\frac{\sqrt{x-1}}{1/2} \right]_t^2 \\
 &= 2 \left(-\lim_{s \rightarrow 1^-} \sqrt{1-s} + \sqrt{1} + \sqrt{1} - \lim_{t \rightarrow 1^+} \sqrt{t-1} \right) = 4
 \end{aligned}$$

$$\begin{aligned}
 \text{(j)} \quad \int_1^\infty \frac{1}{x\sqrt{x^2-1}} dx &= \int_0^\infty \frac{1}{y^2+1} dy \text{ where } y = \sqrt{x^2-1} \\
 &= \lim_{N \rightarrow \infty} \left[\tan^{-1} y \right]_0^N = \lim_{N \rightarrow \infty} \tan^{-1} N - \tan^{-1} 0 = \frac{\pi}{2}
 \end{aligned}$$

6. The **Laplace transform** of a function f , denoted by $\mathcal{L}(f)$, is a function of s defined as

$$\mathcal{L}(f) = \int_0^{\infty} e^{-st} f(t) dt$$

Find the Laplace transform of each of the following functions.



(a) $f(t) = 1$

$$\begin{aligned}\mathcal{L}(1) &= \int_0^{\infty} e^{-st} dt = \lim_{N \rightarrow \infty} \int_0^N e^{-st} dt = \lim_{N \rightarrow \infty} \left[\frac{e^{-st}}{-s} \right]_0^N \\ &= \frac{1}{-s} \left(\lim_{N \rightarrow \infty} \frac{1}{e^{sN}} \right) - \left(\frac{e^0}{-s} \right) = \frac{1}{s} \text{ for } s > 0\end{aligned}$$

(b) $f(t) = e^{at}$

$$\begin{aligned}\mathcal{L}(e^{at}) &= \int_0^{\infty} e^{-st} e^{at} dt = \lim_{N \rightarrow \infty} \int_0^N e^{(a-s)t} dt = \lim_{N \rightarrow \infty} \left[\frac{e^{(a-s)t}}{a-s} \right]_0^N \\ &= \frac{1}{a-s} \left(\lim_{N \rightarrow \infty} \frac{1}{e^{(s-a)N}} \right) - \left(\frac{e^0}{a-s} \right) = \frac{1}{s-a} \text{ for } s > a\end{aligned}$$

(c) $f(t) = t$

$$\begin{aligned}\mathcal{L}(t) &= \int_0^{\infty} e^{-st} t dt = \lim_{N \rightarrow \infty} \int_0^N e^{-st} t dt = \lim_{N \rightarrow \infty} \left[-\frac{e^{-st}}{s^2} - \frac{t e^{-st}}{s} \right]_0^N \\ &= \frac{1}{-s^2} \left(\lim_{N \rightarrow \infty} \frac{1}{e^{sN}} \right) - \frac{1}{s} \left(\lim_{N \rightarrow \infty} \frac{N}{e^{sN}} \right) + \left(\frac{e^0}{s^2} \right) + 0 \\ &= 0 - \frac{1}{s} \left(\lim_{N \rightarrow \infty} \frac{1}{s e^{sN}} \right) + \frac{1}{s^2} = \frac{1}{s^2} \text{ for } s > 0\end{aligned}$$



(d) $f(t) = \sin t$

$$\begin{aligned}\mathcal{L}(\sin t) &= \int_0^{\infty} e^{-st} \sin t dt = \lim_{N \rightarrow \infty} \int_0^N e^{-st} \sin t dt \\ &= \lim_{N \rightarrow \infty} \left[-\frac{e^{-st}(s \sin t + \cos t)}{s^2 + 1} \right]_0^N \\ &= \frac{-1}{s^2 + 1} \left(\lim_{N \rightarrow \infty} \frac{(s \sin N + \cos N)}{e^{sN}} - e^0(s \sin 0 + \cos 0) \right) \\ &= \frac{-1}{s^2 + 1} (0 - 1) = \frac{1}{s^2 + 1} \text{ for } s > 0\end{aligned}$$